Robust Instance-Based Semi-Supervised Learning Change Detection for Remote Sensing Images

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***Abstract*— Semi-supervised change detection (SSCD) has experienced rapid development, with numerous semi-supervised methods being proposed to reduce the reliance on labeled data in change detection. Existing approaches typically rely on manually set high-confidence thresholds to select robust pseudolabels. However, the single-pixel threshold filtering method for pseudolabels (STFPs) lacks context correlation, cannot eliminate high-confidence false positive (HFP) samples, and leads to erroneously filtering out low-confidence true positive (LTP) samples. To address this issue, we propose robust instance-based semi-supervised learning (RISL) change detection for remote sensing (RS) images. RISL evaluates the reliability of each instance object by linking the semantic information of the context, thereby generating robust pseudolabels. In RISL, first, a simple boundary trimming (BT) module as a preprocessing method for change prediction map is introduced. BT can effectively remove low-confidence false positive (LFP) samples while avoiding confusion in the category of instance objects, thereby improving the quality of instance objects. Then, we propose a reliable instance evaluation module (RIEM) to evaluate the reliability of each instance object. RIEM combines the semantic information of the entire instance and establishes correlations between sample contexts to determine the reliability of the instance, effectively eliminating high false positive samples. In addition, consistency regularization (CR) is integrated into RISL, and a new strategy suitable for RIEM is constructed. This strategy enhances the model's generalization ability by mining and hiding semantic information from different views of unlabeled data. Experimental results on the challenging WHU-CD, LEVIR-CD, and CDD-CD datasets show that the proposed method achieves 89.80%, 90.01%, and 87.56% $F1$ scores on labeled data with 5% distribution. RISL achieves state-of-the-art performance compared to other methods.**

***Index Terms*— Instance-based, remote sensing (RS), semi-supervised change detection (SSCD).**

Manuscript received 18 December 2023; revised 6 February 2024; accepted 4 March 2024. Date of publication 20 March 2024; date of current version 28 March 2024. This work was supported in part by the Key Scientific Technological Innovation Research Project by Ministry of Education; in part by the State Key Program and the Foundation for Innovative Research Groups of the National Natural Science Foundation of China under Grant 61836009; in part by the Joint Funds of the National Natural Science Foundation of China under Grant U22B2054; in part by the National Natural Science Foundation of China under Grant 62076192, Grant 62006177, Grant 61902298, Grant 61573267, Grant 61906150, and Grant 62276199; in part by the 111 Project; in part by the Program for Cheung Kong Scholars and Innovative Research Team in University under Grant IRT 15R53; in part by the ST Innovation Project from the Chinese Ministry of Education; in part by the Key Research and Development Program in Shaanxi Province of China under Grant 2019ZDLGY03-06; in part by the National Science Basic Research Plan in Shaanxi Province of China under Grant 2022JQ-607; and in part by the China Postdoctoral fund under Grant 2022T150506. *(Corresponding author: Lingling Li.)*

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Digital Object Identifier 10.1109/TGRS.2024.3379223

I. Introduction

CHANGE detection (CD), as one of the fundamental tasks in remote sensing (RS) interpretation [1], [2], [3], [4], [5], [6], [7], [8], aims to extract change areas from a pair of images taken from high-resolution satellites at the same location but at different times on the ground. CD plays an vital role in disaster response [9], [10], [11], land resources [12], [13], [14], and urban planning [15], [16]. In the past few years, numerous deep neural network-based CD methods [17], [18], [19], [20] have been proposed, achieving impressive results. These methods are capable of mining implicit and abstract feature relationships [21] within CD bi-temporal RS image pairs, leading to a better differentiation of change areas. Although deep neural network-based methods have shown excellent performance in CD, training deep neural networks requires a large amount of labeled data, which will consume a lot of labor and material resources.

Semi-supervised change detection (SSCD) has attracted significant attention for training robust networks by using a small amount of labeled data and mining hidden information from a large amount of unlabeled data. SSCD methods can be broadly divided into adversarial learning-based method [22] and consistency regularization (CR)-based method [23], [24]. In adversarial learning-based methods, two discriminators are used to force the prediction map and pseudolabel entropy map to maintain consistency in feature distribution, thereby improving the generator's generalization ability. In the CR-based method, bi-temporal RS images in different views are forced to generate consistent pseudolabels after passing through the CD model. The method enhances the model's ability to extract semantic information, which can better tackle labeled data lacking challenges. Both directions aim to improve the performance of SSCD models by addressing the challenges posed by limited labeled data and enhancing the robustness of the model in handling diverse environmental conditions. Notable, pseudolabel generation is a very vital part of these methods. Whether in adversarial learning-based methods or CR-based methods, they only use the single-pixel threshold filtering method for pseudolabels (STFP) to generate

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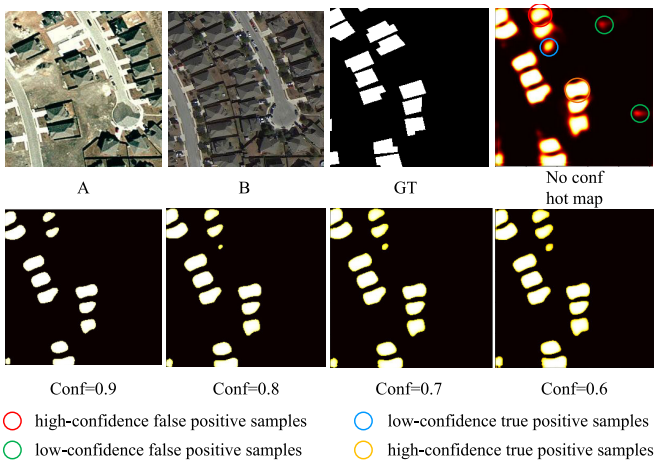


Fig. 1. After training DeepLabv3+ on the LEVIR-CD dataset with 20% labeled data, the inference is performed on the remaining 80% unlabeled data. Multiple confidence thresholds are set in the experiment to check the impact of filtering on pseudolabels. It can be observed that the high-confidence threshold effectively filtered out LFP samples. However, high-confidence thresholds also result in the disappearance of LTP samples and boundary erosion in high-confidence true positives. On the other hand, HFP samples could not be filtered out effectively by high-confidence thresholds.

pseudolabels to supervise the semantic information learning of unlabeled images.

In order to more efficiently utilize unlabeled training data, we conduct a detailed study on pseudolabels [25], [26], [27], [28] generated by the model. Fig. 1 displays the generated pseudolabels by DeepLabv3+ [29] at different confidence thresholds. Through experiments, it is found that at a high-confidence threshold, some low-confidence false positive (LFP) samples [30] can be effectively filtered out. However, as the confidence threshold gradually increases, low-confidence true positive (LTP) samples are erroneously filtered out, but high-confidence false positive (HFPs) samples are still retained.

For the above situation, we figure this is due to the lack of contextual semantic information in STFP. Based on research studies into human thinking processes [1], [31], [32], we have come to understand that when observing the attributes of an object, humans tend to consider the entire instance as a whole. Humans integrate various aspects such as shape, color, texture, and more to arrive at corresponding conclusions.

Therefore, robust instance-based semi-supervised learning (RISL) change detection is proposed, combining the contextual semantic information of each instance object. RISL generates more robust pseudolabels by evaluating and filtering out reliable instance objects, and it can use any type of backbone to replace the CD network. RISL consists of the reliable instance evaluation module (RIEM), the boundary trimming (BT) module, and the CR module. In BT, we use low-confidence threshold to preprocess the change prediction map from the CD network. BT can effectively filter out LFPs, and at the same time, it can effectively trim the edges of each instance to avoid category confusion in instance objects. To evaluate the reliability of the instance object, we designed RIEM. RIEM consists of four components: mask mapping (MM), instances pool (IP), regional judgment selection (RJS), and instance mergence (IM). First, the output of BT is inputted into the MM module. MM utilizes an instance extraction algorithm to segment each instance in the image and maps them to the corresponding regions in the prechange and postchange images to obtain instance mapping image pairs. Then, these instance mapping image pairs are stored in IP, waiting for all instance objects to be segmented. Next, these instance mapping image pairs stored in IP are fed into RJS to determine the reliability of each instance mapping image. Finally, reliable instance mapping image pairs are generated through IM to generate robust pseudolabels. In addition, in order to effectively utilize these unlabeled samples and improve the model's generalization ability under different conditions, RISL utilizes CR to mine rich semantic information hidden in unlabeled data. To begin with, we process the unlabeled image set by applying strong and weak data augmentation to each image, resulting in pairs of augmented samples. Shortly after, these pairs of augmented samples are inputted into the CD network and obtain change prediction maps for each sample pair. Lastly, CR constrains change prediction maps to maintain consistency between pairs of augmented samples.

Overall, we propose RISL in this article. The main contributions can be summarized as follows.

1) The RISL change detection is proposed, which combines RIEM, BT, and CR. Compared with STFP, RISL solves the problem of lacking semantic information in the process of generating pseudolabels from the perspective of instance objects, thereby generating more robust pseudolabels. At the same time, the semantic information hidden in unlabeled data is also mined, improving the semantic extraction ability of the CD network.
2) We propose the RIEM, an effective pseudolabel filtering method. It generates more robust pseudolabels by evaluating the reliability of instances.
3) The BT is designed to remove a large amount of LFP and avoid multiple mixed categories in instance objects. Meanwhile, the consistent regularization (CR) is used to mine rich semantic information from unlabeled samples from different views, thereby maintaining model consistency.
4) Experiments conducted on multiple CD datasets have shown that the RISL can work well on different datasets. It proves the superiority of RISL.

The remaining chapters of this article are arranged as follows. Section II introduces other work related to this article. Section III provides a detailed description of each module in RISL. Section IV presents the experimental results of RISL and other SSCD methods on different datasets, and analyzes their performance and performance reasons. At the same time, ablation experiments were conducted on various modules of RISL. Section V summarizes the entire article and discusses the current limitations and future developments.

Given the extensive use of abbreviations in this article, we have added an abbreviation table (Table I) to assist readers in reading.

II. Related Work

In this section, we introduce fully supervised CD, semantic segmentation with semi-supervision, and SSCD.

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TABLE I
ABBREVIATION TABLE

Abbreviations explained	Abbreviation
The labeled data in the training set	$\mathcal{D}^l$
The pre-change image of labeled samples-pair	$\mathbf{x}_{t1}^i$
The post-change image of labeled samples-pair	$\mathbf{x}_{t2}^i$
The ground-truth of labeled samples-pair	$\mathbf{y}^i$
The unlabeled data in the training set	$\mathcal{D}^u$
The pre-change image of unlabeled samples-pair	$\mathbf{u}_{t1}^j$
The post-change image of unlabeled samples-pair	$\mathbf{u}_{t2}^j$
The CD network	$\mathbf{G}$
The weak augmented view of unlabeled samples-pair	$\left(\mathbf{u}_{t1,w}^j, \mathbf{u}_{t2,w}^j\right)$
The strong augmented view of unlabeled samples-pair	$\left(\mathbf{u}_{t1,s}^j, \mathbf{u}_{t2,s}^j\right)$
The change prediction map of $\left(\mathbf{u}_{t1,w}^j, \mathbf{u}_{t2,w}^j\right)$	$\hat{\mathbf{p}}_w^j$
The change prediction map of $\left(\mathbf{u}_{t1,s}^j, \mathbf{u}_{t2,s}^j\right)$	$\hat{\mathbf{p}}_s^j$
The pseudo-labels after BT	$\hat{\mathbf{p}}_{wt}^j$
The 8-connected regions extracting algorithm	EREA
All instance objects extracted after EREA from $\hat{\mathbf{p}}_{wt}^j$	$\mathcal{O}^j$
The all reliable instance objects obtained from $\hat{\mathbf{p}}_{wt}^j$ after RJS	$\mathcal{C}^j$
The z-th instance mask	$\mathbf{M}_{j,z}$
The robust pseudo-labels	$\hat{\mathbf{p}}_r^j$

A. Fully Supervised CD

Fully supervised CD networks have multiplied in recent years. Fully supervised CD networks can be broadly categorized into CNN-based [33], [34], [35] CD networks and transformer-based [36], [37] CD networks. The CNN-based CD networks focus on local details and can extract texture details of smaller objects within the RS image. Transformer-based CD networks adopt a global perspective and consider the relationships and dependencies between different regions, which can better model the contextual information of RS images.

Daudt et al. [38] built the first end-to-end trained CD network. The siamese network [39], [40] composed of CNN (convolutional neural network) was used to extract the change information. Chen et al. [17] research focused on utilizing attention mechanisms in the CD task. While maintaining the CNN for feature extraction, transformers are combined to model the temporal and spatial domains of paired temporal images. This approach allows for contextual modeling and identifying relevant changes of interest. Bandara and Patel [41] argued that CNNs are unnecessary as a fundamental feature extraction method. Instead, they proposed that transformer encoders can directly and effectively model the context. Therefore, Rumelhart et al. [42] utilized a transformer encoder with a multi-layer perception decoder to aggregate multi-scale information and extract coarse-grained features.

In addition to extracting valuable information from images, modeling time [43], [44] is worth considering. Lin et al. [45] argued that existing CD networks lack decoupling in the temporal dimension, so they focused more on modeling the temporal aspect. They proposed a solution to CD by generating pseudotransition videos from paired temporal images, approaching the problem from a video perspective.

In CD, there is a data imbalance issue between background and foreground classes. This imbalanced sample situation can seriously affect the performance of the network. Feng et al. [46] proposed the dual-branch multilevel intertemporal network to address the issue of imbalanced changes between foreground and background classes.

B. Semantic Segmentation With Semi-Supervision

CD focuses on the semantic information of images, so semi-supervised semantic segmentation can also effectively promote the development of SSCD. In recent years, consistent regularization has been commonly used in semi-supervised semantic segmentation to tap the potential of unlabeled data fully.

French et al. [47] discovered that techniques like CutOut [48] and CutMix [49] have a significant impact on improving CR semi-supervised semantic segmentation methods. Ouali et al. [50] proposed Cross Consistency Training, which leverages unlabeled data by enforcing consistency among images under different perturbations during prediction. Yang et al. [24] reevaluated the consistency framework popularized by FixMatch and proposed a novel augmentation method called auxiliary feature perturbation stream. Additionally, they introduced a dual-stream perturbation technique to fully leverage the augmentation of the original images. Wang et al. [51] proposed a conflict-based cross-view consistency (CCVC) strategy, which involves collaborative training by learning reliable predictions from two unrelated views. They also introduced the conflict-based pseudolabeling (CPL) method to enhance the learning of useful semantic information and stabilize the training process.

In addition to applying augmentation transformations to the overall image to enforce consistency, Lai et al. [52] proposed that despite the unknown ground-truth (GT) labels in different environments, a pixel-wise correspondence exists between the overlapping regions of the two crops. This enables the preservation of high-level feature consistency. They introduced the directional contrastive loss to address overfitting issues [53] and maintain contextual awareness between pixels in different environments. This loss enhances the robustness of the model and promotes pixel-level consistency.

C. Semi-Supervised Change Detection

Although most fully supervised methods have achieved impressive performance in CD, they require a large amount of labeled data to learn. In contrast, SSCD requires fewer labeled data than fully supervised approaches. They leverage information from vast unlabeled data to extract effective features.

Peng et al. [22] proposed a semi-supervised convolutional network based on GAN. By employing two discriminators, the segmentation maps and entropy maps distributions of labeled and unlabeled data are aligned, allowing for effective utilization of unlabeled data. Indeed, methods based on adversarial learning can be challenging to optimize [54].

CD and semantic segmentation networks share similarities in their structural design, as both require high-level semantic segmentation of objects in the image. Therefore, the ideas and techniques from semi-supervised semantic segmentation methods can also be leveraged in SSCD. The approach of enhancing the consistency of the model across different views based on weak-strong augmentations has been incorporated into SSCD methods.

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Bandara and Patel [41] argued that the clustering assumption holds for deep feature difference-domain in CD and then proposed a semi-supervised CD paradigm with CR. In this paradigm, small perturbations are added to the bi-temporal image pair, and the predictions with perturbations are constrained to be consistent with the predictions without perturbations. Zhang et al. [55] proposed the feature-prediction alignment framework for SSCD. FPA utilizes two strategies to constrain the training of the model. First, it employs the feature-prediction alignment strategy, which leverages class-aware feature alignment to reduce feature discrepancies among samples of the same class. Second, FPA applies strong and weak augmentations to the data and utilizes pixel-wise prediction alignment to align the pseudolabels generated by the two augmentations. It enhances the model's adaptability to different perspectives.

III. Methodology

In this section, the details of each module of RISL are introduced in detail. Section III-A describes the symbol definition of the dataset and CD network. Section III-B introduces the use of weakly strong view CR in this article. Section III-C introduces BT. Afterward, in Section III-D, the specific content of each module of RIEM is described in detail. Finally, in Section III-E, the loss constraint function used in RISL is introduced. Fig. 2 shows the overall structure of RISL.

A. Problem Definition

In SSCD, the labeled data in the training set are defined as $\mathcal{D}^l = \{(\mathbf{x}_{t1}^i, \mathbf{x}_{t2}^i, \mathbf{y}^i)\}_{i=1}^M$. $(\mathbf{x}_{t1}^i, \mathbf{x}_{t2}^i, \mathbf{y}^i)$ represents the ith pair of labeled samples-pair. In the ith pair of labeled samples-pair $\mathbf{x}_{t1}^i \in \mathbb{R}^{3\times W\times H}$ represents the prechange image, $\mathbf{x}_{t2}^i \in \mathbb{R}^{3\times W\times H}$ represents the postchange image, and $\mathbf{y}^i \in \mathbb{R}^{1\times W\times H}$ represents the corresponding GT mask. M represents the number of labeled sample pairs. We use $\mathcal{D}^u = \{(\mathbf{u}_{t1}^j, \mathbf{u}_{t2}^j)\}_{j=1}^N$ to represent the unlabeled training data. $(\mathbf{u}_{t1}^j, \mathbf{u}_{t2}^j)$ represents the jth unlabeled sample pairs, $\mathbf{u}_{t1}^j \in \mathbb{R}^{3\times W\times H}$ represents the prechange image, and $\mathbf{u}_{t2}^j \in \mathbb{R}^{3\times W\times H}$ represents the postchange image in the jth pair of unlabeled samples-pair. N represents the number of unlabeled sample pairs. In addition, the CD network is defined as $\mathbf{G}$.

B. CR of Weak-Strong Views

In semi-supervised learning, CR is an essential and effective method to explore the rich information embedded in unlabeled data. To adapt to RSCD tasks, we modify the CR methods used in semantic segmentation.

First, weak data augmentations and strong data augmentations are applied to the unlabeled samples-pair $(\mathbf{u}_{t1}^j, \mathbf{u}_{t2}^j)$, creating two pairs of samples with different views by creating weak augmented view $(\mathbf{u}_{t1,w}^j, \mathbf{u}_{t2,w}^j)$ and strong augmented view $(\mathbf{u}_{t1,s}^j, \mathbf{u}_{t2,s}^j)$. Then, different view pairs $(\mathbf{u}_{t1,w}^j, \mathbf{u}_{t2,w}^j)$ and $(\mathbf{u}_{t1,s}^j, \mathbf{u}_{t2,s}^j)$ are inputted to model $\mathbf{G}$, obtaining two change prediction maps, $\hat{p}_w^j \in \mathbb{R}^{1\times W\times H}$ and $\hat{p}_s^j \in \mathbb{R}^{1\times W\times H}$, corresponding to the different views. Finally, we utilize Cross-Entropy [56] to enforce consistency between the two views, which can be formulated as

$$\hat{p}_w^j = \mathbf{G}\left(\mathbf{u}_{t1,w}^j, \mathbf{u}_{t2,w}^j\right) \tag{1}$$

$$\hat{p}_s^j = \mathbf{G}\left(\mathbf{u}_{t1,s}^j, \mathbf{u}_{t2,s}^j\right) \tag{2}$$

$$\mathcal{L}_u = \mathrm{CE}\left(\hat{p}_w^j, \hat{p}_s^j\right) \tag{3}$$

where $\mathrm{CE}(\cdot)$ represents the Cross-Entropy.

C. Boundary Trimming

Due to the limited quantity of labeled data, the predictions of model $\mathbf{G}$ may be inaccurate. If instance extraction is directly performed on the prediction of model $\mathbf{G}$, it may result in a significant number of LFPs being retained. It may also lead to confusion in instance categories. For example, an instance belonging to changed instance may contain too many pixels from unchanged instance. It can also result in multiple instances of different categories being connected and treated as a single instance with a mixed category.

Fortunately, we observe that using a low-confidence threshold can effectively filter out LFPs and accurately segment the boundaries of each instance. Therefore, a low-confidence-based BT module is proposed. BT takes the output $\hat{p}_w^j$ of model $\mathbf{G}$ as input. First, $\hat{p}_w^j$ is normalized using softmax [57]. Then, a threshold value τ is applied to filter out sample pixels with predicted class probability greater than τ, resulting in the trimmed pseudolabels $\hat{p}_{wt}^j$, which can be formulated as

$$\hat{p}_{wt}^{j,m} = \begin{cases} 0 & \text{if } \hat{p}_{wt}^{j,m} < \tau \\ 1 & \text{else} \end{cases} \tag{4}$$

where $\hat{p}_{wt}^{j,m}$ represents the label of the mth pixel in the trimmed pseudolabels $\hat{p}_{wt}^j$. $\hat{p}_{wt}^j$ represents the jth label in the pseudolabels of model $\mathbf{G}$ after Softmax. Fig. 3 shows the overall structure of BT.

D. Reliable Instance Evaluation Module

Most semi-supervised algorithms utilize STFP to obtain more robust pseudolabels. As shown in Fig. 1, although high-confidence threshold can eliminate LFPs, it fails to remove HFPs and may incorrectly filter out some LTPs. From a perception perspective, humans judge positive or negative samples based on the entire instance object rather than single pixel. This instance-based evaluation approach allows for careful consideration of multiple aspects such as color, shape, and size, making it more comprehensive and stable than methods based on STFP. Therefore, to address this issue, we propose the RIEM (Fig. 4), which evaluates the reliability of each instance object and performs instance-level filtering to obtain robust pseudolabels. RIEM has been illustrated in Algorithm 1.

In RIEM, the first step is MM. The eight-connected regions extracting algorithm (EREA) is used to extract all instance objects in $\hat{p}_{wt}^j$.

In the EREA, the foreground is non-zero pixels, while the background is zero pixels. EREA traverses each pixel and, starting from the current pixel recursively searches its eight neighboring pixels. If the adjacent pixel is a foreground pixel

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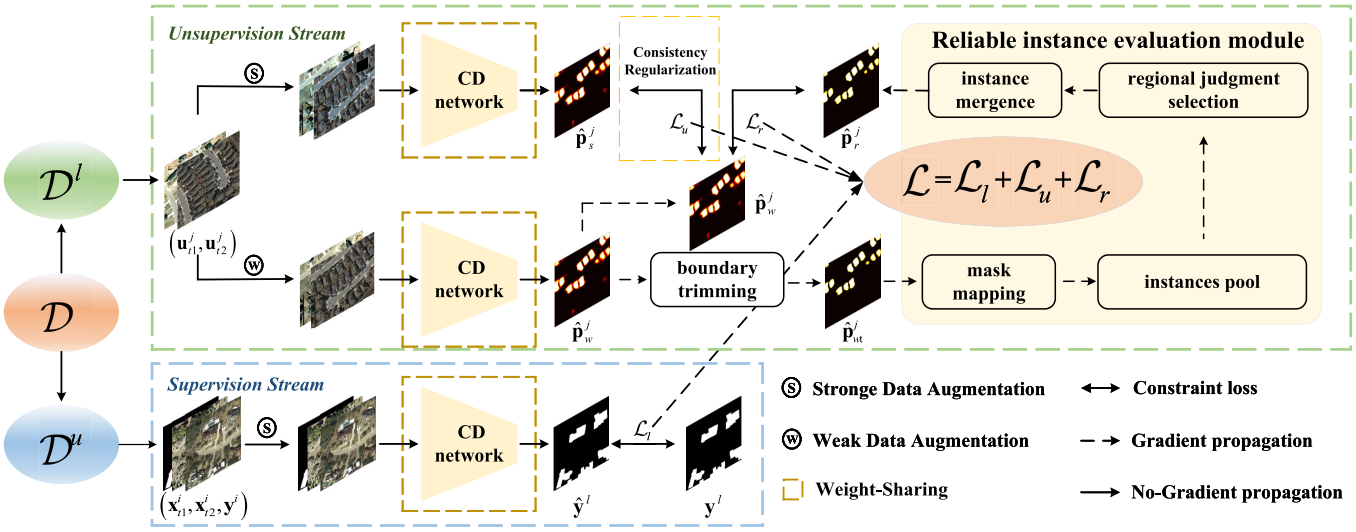


Fig. 2. Overall architecture of RISL change detection. Simultaneously applying labeled sample pairs $(\mathbf{x}_{t1}^i, \mathbf{x}_{t2}^i)$ and unlabeled sample pairs $(\mathbf{u}_{t1}^j, \mathbf{u}_{t2}^j)$ in a batch, respectively, to the supervision stream and the unsupervision stream. in the supervision stream, conducting conventional supervised training. In the unsupervision stream, first obtaining weakly augmented pseudolabels, then pruning the edges through BT, followed by RIEM to filter reliable instance objects and merge them into robust pseudolabels. Finally, training is performed by computing the loss using the obtained robust pseudolabels along with strongly augmented predictions.

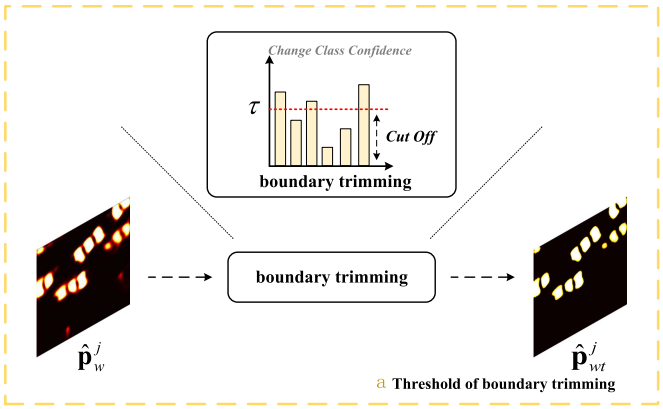


Fig. 3. Overall structure of BT. Predicted values below τ will be cut off.

and has not been marked, it will be marked and belong to the same connected area. If the current pixel has already been marked, it will be skipped. Multiple connected regions are ultimately obtained through repeated traversal, and each region is extracted as an instance object. It can be represented as follows:

$$\mathcal{O}^j = \{\mathbf{O}_1, \mathbf{O}_2, \ldots, \mathbf{O}_{t_{\max}}\} = \mathrm{EREA}\left(\hat{p}_{wt}^j\right) \tag{5}$$

where $\mathrm{EREA}(\cdot)$ represents the eight-connected regions extracting algorithm. $\mathcal{O}^j$ represents the collection of all instance objects extracted from $\hat{p}_{wt}^j$. $\mathbf{O}_t \in \mathbb{R}^{1\times W_c\times H_c}$ represents the tth instance object. $t_{\max}$ represents the number of extracting instances.

Then, we crop the prechange image and postchange image $(\mathbf{u}_{t1,w}^j, \mathbf{u}_{t2,w}^j)$ based on the extracted instance objects $\mathbf{O}_t$ to obtain the corresponding instance mapping image pair with respect to $\mathbf{O}_t$. It can be represented as follows:

$$\left(\mathbf{O}_{j,t}^{t1}, \mathbf{O}_{j,t}^{t2}\right) = \mathrm{MaskMapping}\left(\mathbf{O}_t, \left(\mathbf{u}_{t1,w}^j, \mathbf{u}_{t2,w}^j\right)\right) \tag{6}$$

$$\mathbf{IP} \leftarrow \left(\mathbf{O}_{j,t}^{t1}, \mathbf{O}_{j,t}^{t2}\right) \tag{7}$$

Algorithm 1 Pseudocode for RIEM

Input: the output $\hat{p}_{wt}^j$ of BT, the weak augmented view $(\mathbf{u}_{t1,w}^j, \mathbf{u}_{t2,w}^j)$, a robust pretrain model RJM.

Output: reliable pseudolabels $\hat{p}_r^j$.

```
Get all instance objects: O^j=EREA(p̂^j_wt);
for O_t ∈ O^j do
    Obtain corresponding instance mapping image pair:
      (O^t1_{j,t}, O^t2_{j,t}) = MaskMapping(O_t, (u^j_{t1,w}, u^j_{t2,w}));
    IP ← (O^t1_{j,t}, O^t2_{j,t});
end
for (O^t1_{j,t}, O^t2_{j,t}) ∈ IP do
    Evaluate whether the image pair of the instance
      object represents a changed instance:
      C_{j,t} = RJM((O^t1_{j,t}, O^t2_{j,t}));
    if C_{j,t} == 1 then
        changed instance: C^j ← C_{j,t};
    else
        unchanged instance: C^j ⊀ C_{j,t};
    end
end
Get robust pseudolabels: p̂^j_r =IM(C^j, O^j);
```

where $(\mathbf{O}_{j,t}^{t1}, \mathbf{O}_{j,t}^{t2})$ represents the instance objects $\mathbf{O}_t$ corresponding to the image regions in $(\mathbf{u}_{t1,w}^j, \mathbf{u}_{t2,w}^j)$. $\mathbf{O}_{j,t}^{t1} \in \mathbb{R}^{3\times W_c\times H_c}$ and $\mathbf{O}_{j,t}^{t2} \in \mathbb{R}^{3\times W_c\times H_c}$. **IP** represents the set of $(\mathbf{O}_{j,t}^{t1}, \mathbf{O}_{j,t}^{t2})$.

Afterward, add $(\mathbf{O}_{j,t}^{t1}, \mathbf{O}_{j,t}^{t2})$ into the RJS to determine whether the image pair of the instance object represents a changed instance, as shown in (8) and (9). For RJS, a robust model called the region judgment model (RJM) needs to be pretrained to identify changed instances and unchanged

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Algorithm 2 Pseudocode of RJM Pretraining

Input: A classification model RJM, the ith pair of labeled training data $(\mathbf{x}_{t1}^i, \mathbf{x}_{t2}^i)$, optimizer AdamW.

Output: a robust pretrain model RJM.

Extract individual connected domains with pixels greater than 0 in $\mathbf{y}^i$ and obtain the matrix boxes for each connected domain: $(X, Y, W, H) = Extract(\mathbf{y}^i)$;
for $(x, y, w, h) \in (X, Y, W, H)$ **do**
  Cut out the corresponding areas on $(\mathbf{x}_{t1}^i, \mathbf{x}_{t2}^i)$ as positive samples: $\mathbf{q}_{t1}^{pos}$=$Crop(\mathbf{x}_{t1}^i, (x, y, w, h))$, $\mathbf{q}_{t2}^{pos}$=$Crop(\mathbf{x}_{t2}^i, (x, y, w, h))$, $\mathbf{l}^{pos} = 1$;
  $(\mathcal{Q}_{t1}, \mathcal{Q}_{t2}, \mathcal{QT}) \leftarrow (\mathbf{q}_{t1}^{pos}, \mathbf{q}_{t2}^{pos}, \mathbf{l}^{pos})$;
  $w_{neg} = w, h_{neg} = h, n = 0$;
  while $n \leq 3$ **do**
    Randomly crop rectangles of sizes $w_{neg} \times h_{neg}$ in $\mathbf{y}^i$. $(x_{neg}, y_{neg}, w_{neg}, h_{neg}) = RandomlyCrop(\mathbf{y}^i, (w_{neg}, h_{neg}))$;
    if $(x_{neg}, y_{neg}, w_{neg}, h_{neg})$ *not in* (X, Y, W, H) **then**
      Cut out the corresponding areas on $(\mathbf{x}_{t1}^i, \mathbf{x}_{t2}^i)$ as negative samples: $\mathbf{q}_{t1}^{pos}$=$Crop(\mathbf{x}_{t1}^i, (x_{neg}, y_{neg}, w_{neg}, h_{neg}))$, $\mathbf{q}_{t2}^{pos}$=$Crop(\mathbf{x}_{t2}^i, (x_{neg}, y_{neg}, w_{neg}, h_{neg}))$, $\mathbf{l}^{neg} = 0$;
      Resize image size to 128×128, $\mathbf{q}_{t1}^{pos}$=$Resize(\mathbf{q}_{t1}^{pos}, (128, 128))$;
      Resize image size to 128×128, $\mathbf{q}_{t2}^{pos}$=$Resize(\mathbf{q}_{t2}^{pos}, (128, 128))$; $(\mathcal{Q}_{t1}, \mathcal{Q}_{t2}, \mathcal{QT}) \leftarrow (\mathbf{q}_{t1}^{neg}, \mathbf{q}_{t2}^{neg}, \mathbf{l}^{neg})$;
      Break;
    end
    $w_{neg} = \times 3/2, h_{neg} = \times 3/2, n = n + 1$;
  end
end
for $(\mathbf{q}_{t1}^c, \mathbf{q}_{t2}^c, \mathbf{l}^c) \in (\mathcal{Q}_{t1}, \mathcal{Q}_{t2}, \mathcal{QT})$ **do**
  $class^c$ = RJM$(\mathbf{q}_{t1}^c, \mathbf{q}_{t2}^c)$;
  Calculate RJM loss: loss = CE$(class^c, \mathbf{l}^c)$;
  AdamW(loss);
  Update RJM weights: update(RJM);
end

instances in the image. RJM is defined as a neural network model with binary classification capability. In this article, RJM uses the Resnet-18 network as the backbone, concatenates the obtained feature maps together, passes through an average pooling layer, and finally uses a fully connected layer for classification. The specific training procedure can be referred to as Algorithm 2. Algorithm 2 cuts out positive and negative samples from labeled data $(\mathbf{x}_{t1}^i, \mathbf{x}_{t2}^i)$. The label of the negative sample is 0, indicating the unchanged instance in the image pair. The positive sample class label is 1, indicating the changed instance of the image pair. Then, resize the instances to 128×128 and use them as data for training RJM. Finally,

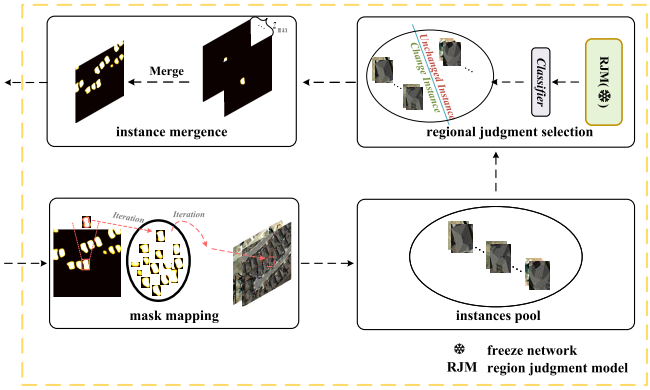


Fig. 4. Overall architecture of RIEM. First, use the EREA algorithm to extract instance objects and map the extracted objects to prechange and postchange images to obtain instance mapping image pairs. Then, input instance mapping image pairs into RJM to evaluate the reliability of the instance. Finally, merge all reliable instances to obtain robust pseudolabels.

use the resized data to train RJM

$$C_{j,t} = \mathrm{RJM}\left(\left(\mathbf{O}_{j,t}^{t1}, \mathbf{O}_{j,t}^{t2}\right)\right) \tag{8}$$

$$\mathcal{C}^j = \begin{cases} \mathcal{C}^j \leftarrow C_{j,t} & \text{if } C_{j,t} = 1 \\ \mathcal{C}^j \not\leftarrow C_{j,t} & \text{else} \end{cases} \tag{9}$$

where $\mathcal{C}^j \in \{C_{j,0}, C_{j,1}, \ldots, C_{j,z_{\max}}\}$ represents all reliable instance objects obtained from $\hat{p}_{wt}^j$ after being filtered through RJS. $z_{\max}$ represents the number of $\mathcal{C}^j$. $C_{j,z} \in \mathbb{R}^{1\times W_1 \times H_1}$ represents zth reliable instance object in $\mathcal{C}^j$.

Then traverse $C_{j,t}$ in $\mathcal{C}^j$, because each $C_{j,t}$ corresponds to a $\mathbf{O}_t$, so IM overlaps these $\mathbf{O}_t$ and combines them into the final robust pseudolabels $\hat{p}_r^j \in \mathbb{R}^{1\times W\times H}$.

Finally, we utilize cross-entropy as a constraint as follows to bring $\hat{p}_r^j$ closer to $\hat{p}_w^j$, enabling the maximization of the potential of unlabeled data to the greatest extent possible:

$$\mathcal{L}_r = \mathrm{CE}\left(\hat{p}_r^j, \hat{p}_w^j\right). \tag{10}$$

E. Loss Function

The loss function of RISL consists of three components: fully supervised loss for labeled data, CR loss (3), and RIEM loss (10). Their mathematical representations are as follows:

$$\mathcal{L} = \mathcal{L}_l + \mathcal{L}_u + \mathcal{L}_r \tag{11}$$

where $\mathcal{L}_l$ represents the supervised training loss on labeled data, which can be represented mathematically as follows:

$$\mathcal{L}_l = \mathrm{CE}\left(\hat{y}^l, \mathbf{y}^l\right). \tag{12}$$

IV. Experiments

A. Datasets

We conducted experiments on the LEVIR-CD, WHU-CD, and CDD-CD datasets to validate the effectiveness of RISL.

1) *LEVIR-CD [60]:* The LEVIR-CD dataset consists of image pairs from 20 different regions in multiple cities in Texas, United States. It comprises 637 pairs of high-resolution Google Earth (GE) captured images, each with a size of

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TABLE II

COMPARATIVE EXPERIMENTAL RESULTS OF DIFFERENT CD METHODS ON THE WHU-CD, LEVIR-CD AND CDD-CD DATASET. EXCEPT FOR KAPPA, ALL SCORES ARE EXPRESSED IN PERCENTAGE (%). EXCEPT FOR FULL-SUP, THE HIGHEST SCORE IS MARKED IN **BOLD**

Method			AdvEnt [58]	S4GAN [59]	SemiCDNet [22]	SemiCD [23]	UniMatch [24]	ISCDNet	Only-sup	Full-sup
WHU-CD	5%	Pre	71.82	64.14	73.79	79.60	88.10	**89.24**	56.57	93.95
		Rec	86.88	88.29	90.96	79.13	89.12	90.37	**94.60**	94.34
		F1	78.63	74.30	81.48	79.36	88.61	**89.80**	70.79	94.15
		IOU	64.79	59.11	68.75	65.79	79.54	**81.48**	54.79	88.94
		Kappa	0.7766	0.7306	0.8064	0.7851	0.8813	**0.8932**	0.6886	0.8894
	10%	Pre	75.87	68.27	76.18	79.09	**89.35**	87.65	65.37	93.95
		Rec	90.32	88.49	93.09	82.95	90.04	**93.46**	91.87	94.34
		F1	82.47	77.07	83.79	80.97	89.69	**90.46**	76.39	94.15
		IOU	70.17	62.70	72.11	68.03	81.31	**82.59**	61.79	88.94
		Kappa	0.8168	0.7600	0.8305	0.8017	0.8926	**0.8995**	0.7494	0.8894
	20%	Pre	**96.08**	80.85	84.61	84.67	89.47	91.62	70.78	93.95
		Rec	76.64	84.22	86.04	86.19	90.57	90.66	**93.29**	94.34
		F1	85.26	82.50	85.32	85.42	90.01	**91.14**	80.49	94.15
		IOU	74.31	70.22	74.40	74.55	81.84	**83.72**	67.35	88.94
		Kappa	0.8473	0.8177	0.8471	0.8481	0.8960	**0.9068**	0.7933	0.8894
	40%	Pre	89.44	83.00	82.09	86.67	90.83	89.71	**92.03**	93.95
		Rec	84.08	88.34	92.24	88.65	92.72	**94.4**	83.84	94.34
		F1	86.68	85.59	86.87	87.65	91.77	**92.00**	87.75	94.15
		IOU	76.48	74.81	76.79	78.01	84.79	**85.18**	78.16	88.94
		Kappa	0.8615	0.8497	0.8630	0.8713	0.9142	**0.9157**	0.8714	0.8894
LEVIR-CD	5%	Pre	92.89	89.25	87.53	91.37	**96.09**	94.71	88.57	92.34
		Rec	74.98	69.92	78.35	79.79	83.63	**85.76**	51.38	86.74
		F1	82.98	78.41	82.69	85.18	89.43	**90.01**	65.03	89.45
		IOU	70.91	64.49	70.48	74.19	80.88	**81.84**	48.18	80.91
		Kappa	0.8217	0.7740	0.8181	0.8445	0.8890	**0.8949**	0.6367	0.8890
	10%	Pre	90.10	91.58	89.85	91.42	**96.62**	92.11	92.53	92.34
		Rec	79.97	70.28	80.64	83.13	84.14	**88.59**	76.89	86.74
		F1	84.74	79.53	85.00	87.08	89.95	**90.32**	83.99	89.45
		IOU	73.52	66.02	73.91	77.12	81.73	**82.34**	72.39	80.91
		Kappa	0.8397	0.7859	0.8424	0.8642	0.8895	**0.8981**	0.8321	0.8890
	20%	Pre	89.12	89.09	89.41	91.75	**95.07**	93.93	92.90	92.34
		Rec	81,06	79.70	81.23	83.71	85.68	**87.18**	81.62	86.74
		F1	84.90	84.14	85.12	87.55	90.13	**90.43**	86.89	89.45
		IOU	73.77	72.62	74.10	77.85	82.04	**82.53**	76.83	80.91
		Kappa	0.8413	0.8333	0.8436	0.8691	0.8963	**0.8992**	0.8624	0.8890
	40%	Pre	89.13	89.68	88.49	90.83	**95.42**	93.15	92.97	92.34
		Rec	81.23	80.84	82.49	85.84	85.63	**89.06**	82.64	86.74
		F1	85.00	85.03	85.38	88.26	90.26	**91.06**	87.50	89.45
		IOU	73.91	73.96	74.50	78.99	82.25	**83.58**	77.77	80.91
		Kappa	0.8423	0.8427	0.8463	0.8765	0.8977	**0.9059**	0.8687	0.8890
CDD	5%	Pre	87.56	76.61	84.51	81.22	**91.39**	88.23	81.64	95.84
		Rec	62.50	71.88	70.29	75.48	79.57	**86.90**	70.37	91.34
		F1	72.94	74.16	76.74	78.24	85.08	**87.56**	75.59	93.54
		IOU	57.40	58.94	62.27	64.26	74.03	**77.47**	60.75	87.86
		Kappa	0.6967	0.7047	0.7366	0.7515	0.8324	**0.8589**	0.7259	0.9270
	10%	Pre	89.72	85.91	89.27	84.69	**93.63**	93.25	86.72	95.84
		Rec	72.15	78.17	78.29	85.29	86.05	**88.95**	74.82	91.34
		F1	79.98	81.86	83.42	84.99	89.68	**91.05**	80.34	93.54
		IOU	66.64	69.29	71.56	73.90	81.29	**83.57**	67.13	87.86
		Kappa	0.7737	0.7931	0.8114	0.8275	0.8837	**0.8988**	0.7792	0.9270
	20%	Pre	94.25	92.96	91.92	89.57	**94.80**	93.81	92.76	95.84
		Rec	78.72	82.42	84.29	88.61	90.58	**93.87**	88.30	91.34
		F1	85.79	87.37	87.94	89.09	92.64	**93.84**	90.48	93.54
		IOU	75.12	77.58	78.47	80.32	86.29	**88.39**	82.61	87.86
		Kappa	0.8390	0.8563	0.8624	0.8748	0.9168	**0.9302**	0.8924	0.9270
	40%	Pre	92.36	93.49	92.72	91.93	**95.80**	94.98	94.72	95.84
		Rec	86.11	86.86	86.62	89.64	93.17	**94.87**	83.78	91.34
		F1	89.13	90.06	89.57	90.77	94.46	**94.92**	88.91	93.54
		IOU	80.39	81.91	81.10	83.10	89.51	**90.34**	80.03	87.86
		Kappa	0.8758	0.8864	0.8808	0.8942	0.9373	**0.9425**	0.8753	0.9270

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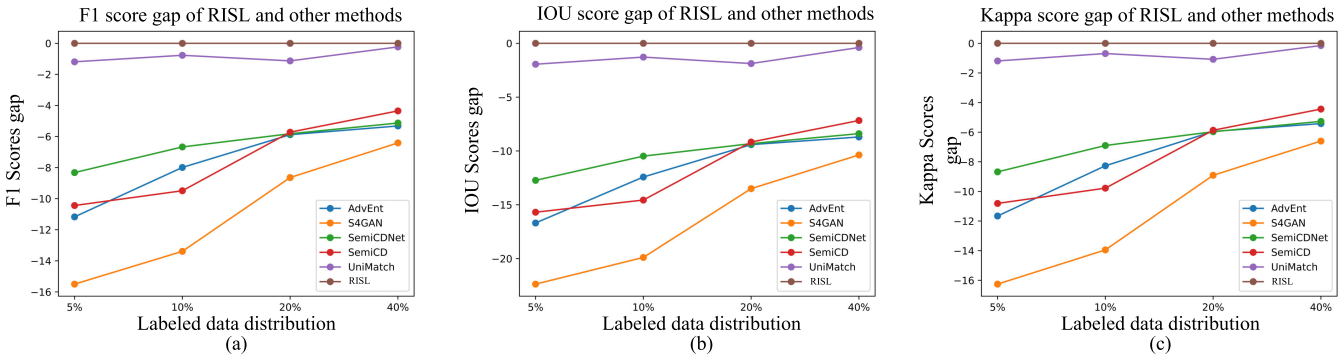


Fig. 5. Gap between the proposed RISL and other comparison methods on the WHU-CD dataset. (a) Gap in $F1$. (b) Gap in IOU. (c) Gap in Kappa.

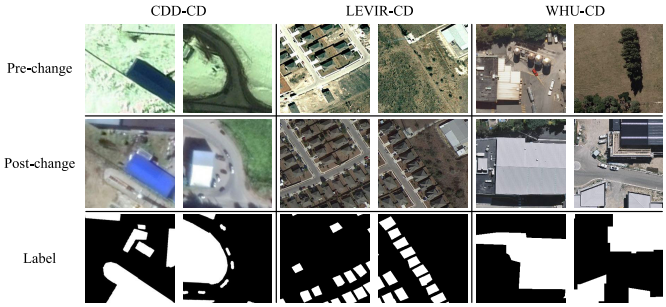


Fig. 6. Some examples of LEVIR-CD, WHU-CD, and CDD-CD datasets.

1024 × 1024 pixels. In the experiments, these image patches are further split into smaller patches of size 256 × 256, resulting in a training/validation/test split of 7120/1024/2048 image pairs, respectively.

2) *WHU-CD [61]:* The WHU-CD dataset consists of aerial images captured in the same area of Christchurch, New Zealand, between 2012 and 2016. Each image has a size of 32 507 × 15 345 pixels and a resolution of 0.075 m. In the experiments, these images are split into smaller patches of size 256 × 256. The dataset is then divided into training (80%), validation (10%), and testing (10%) sets following the specified proportions.

3) *CDD-CD [62]:* The CDD-CD dataset consists of seasonal RS images of the same area captured by GE. The images obtained have a size of 256 × 256 pixels and a spatial resolution ranging from 3 to 100 cm/px. CDD contains a diverse range of changing objects, including cars, large building structures, and seasonal variations in trees, among others. The dataset is divided into three subsets for training, validation, and testing, with 10 000, 3000, and 3000 images, respectively. Table II shows the results on three datasets. Fig. 5 shows gap between RISL and other methods. Fig. 6 shows some examples of three datasets.

B. Evaluation Metrics

In order to assess the performance of RISL from multiple aspects, we introduce five CD evaluation metrics: precision (Pre), recall (Rec), $F1$-score ($F1$), intersection over union (IOU), and kappa coefficient (Kappa). Among them, the metric IOU is the change IOU. These metrics aim to provide a comprehensive evaluation of the model's performance. The four metrics are formulated as

$$\mathbf{Pre} = \frac{\mathbf{TP}}{\mathbf{TP}+\mathbf{FP}} \times \mathbf{100\%} \tag{13}$$

$$\mathbf{Rec} = \frac{\mathbf{TP}}{\mathbf{TP}+\mathbf{FN}} \times \mathbf{100\%} \tag{14}$$

$$\mathbf{F1} = \mathbf{2} \times \frac{\mathbf{Pre} \times \mathbf{Rec}}{\mathbf{Pre}+\mathbf{Rec}} \times \mathbf{100\%} \tag{15}$$

$$\mathbf{IOU} = \frac{\mathbf{TP}}{\mathbf{TP}+\mathbf{FP}+\mathbf{FN}} \times \mathbf{100\%} \tag{16}$$

$$\mathbf{OA} = \frac{\mathbf{TP}+\mathbf{TN}}{\mathbf{TP}+\mathbf{TN}+\mathbf{FP}+\mathbf{FN}} \times \mathbf{100\%} \tag{17}$$

$$\mathbf{Pe} = \mathbf{100\%} \times \left(\frac{(\mathbf{TP}+\mathbf{FP}) \times (\mathbf{TP}+\mathbf{FN})}{(\mathbf{TP}+\mathbf{TN}+\mathbf{FP}+\mathbf{FN})^2} + \frac{(\mathbf{FP}+\mathbf{TN}) \times (\mathbf{FN}+\mathbf{TN})}{(\mathbf{TP}+\mathbf{TN}+\mathbf{FP}+\mathbf{FN})^2} \right) \tag{18}$$

$$\mathbf{Kappa} = \frac{\mathbf{OA}-\mathbf{Pe}}{\mathbf{1}-\mathbf{Pe}} \tag{19}$$

where true positive (TP) represents the number of pixels correctly detected as changed. False positive (FP) represents the number of pixels incorrectly detected as changed. True negative (TN) represents the number of pixels correctly detected as no-changed. False negative (FN) represents the number of pixels incorrectly detected as no-changed. The range of the five evaluation metrics, **Pre**, **Rec**, **F1** and **IOU**, is from 0% to 100%. The range of the **Kappa** is −1 to 1.

C. Experimental Settings

To ensure a fair comparison with previous work, we adopt DeepLabv3+ (backbone is Resnet-50, as shown in Fig. 7) as the CD network **G**. In addition, we also utilize the SGD optimizer [63] to update the network's parameters. In **G**, to adapt to the RS change detection, we calculate the feature difference between stage 1 and stage 4 and fuse the features of the two stages to ensure the network can adapt to both large and small target change areas simultaneously. The structure of stages 1, 2, 3, and 4 in RISL is the same as that of stages 1, 2, 3, and 4 in Resnet-50.

We added sensitivity experiments for τ (as shown in Tables III and IV) to demonstrate the performance change in RISL by varying the threshold value. From the table, it can be seen that RISL performs the best when t is 0.7. From the two tables, we can see that RISL performs best when τ is 0.7.

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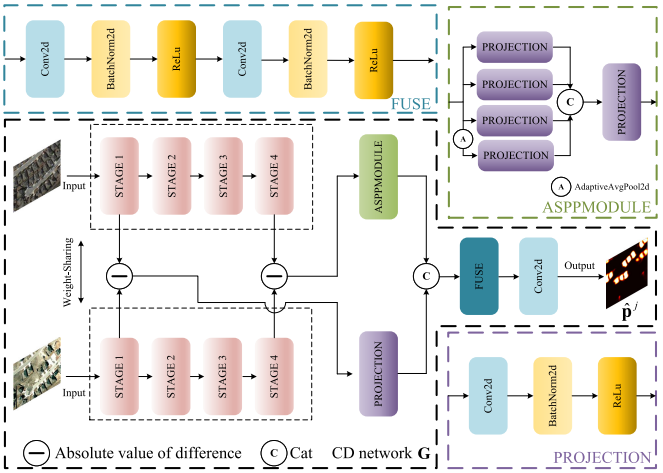


Fig. 7. Detailed structure of CD network **G**. The Stage 1, Stage 2, Stage 3, and Stage 4 are Stage 1, Stage 2, Stage 3, and Stage 4 in the Resnet-50 structure, respectively.

TABLE III

PERFORMANCE OF RISL WITH DIFFERENT THRESHOLDS τ ON 20% LABELED DATA IN WHU-CD. EXCEPT FOR KAPPA, ALL SCORES ARE EXPRESSED IN PERCENTAGE (%)

τ	0.6	0.7	0.8	0.9
Pre	93.78	91.62	94.96	95.26
Rec	88.02	90.66	86.08	84.13
F1	90.81	91.14	90.30	89.36
IOU	83.17	83.72	82.32	80.76
Kappa	0.9044	0.9068	0.8992	0.8894

TABLE IV

PERFORMANCE OF RISL WITH DIFFERENT THRESHOLDS τ ON 5% LABELED DATA IN CDD-CD. EXCEPT FOR KAPPA, ALL SCORES ARE EXPRESSED IN PERCENTAGE (%)

τ	0.6	0.7	0.8	0.9
Pre	93.24	88.23	95.04	96.70
Rec	81.16	86.90	80.52	76.02
F1	86.78	87.56	87.18	85.12
IOU	76.65	77.87	77.28	74.10
Kappa	0.8515	0.8589	0.8563	0.8340

The hyperparameters for SGD include a weight decay of 0.0001, momentum of 0.9, and an initial learning rate of 0.02. The learning rate is updated through (20). All models are implemented using PyTorch [64]. The batch size is set to 16, where each batch consists of an equal combination of labeled and unlabeled data. Each model is trained for 100 epochs

$$lr = lr_b \times \left(1 - \text{epoch}_c/\text{epoch}_m\right)^{0.9} \tag{20}$$

where lr_b refers to the initial learning rate. epoch_c represents the current number of epochs completed. epoch_m denotes the maximum number of epochs for training iteration. lr represents the current learning rate.

To fully leverage the potential of unlabeled data, we employ a weak augmentation strategy consisting of Random Crop, Random Horizontal Flip, Random Vertical Flip, and Random Rotate.

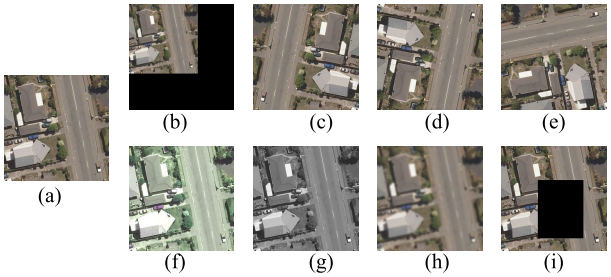


Fig. 8. Impact of data augmentation strategies on images. (a) Original. (b) Random Crop. (c) Random Horizontal Flip. (d) Random Vertical Flip. (e) Random Rotate. (f) ColorJitter. (g) RandomGrayscale. (h) Blur. (i) CutOut.

TABLE V

RISL'S ABLATION RESULTS ON THE 20% LABELED LEVIR-CD AND WHU-CD DATASETS. EXCEPT FOR KAPPA, ALL SCORES ARE EXPRESSED IN PERCENTAGE (%)

Method	LEVIR-CD			WHU-CD		
	F1	IOU	Kappa	F1	IOU	Kappa
The effectiveness of CR						
Component 1	86.89	76.83	0.8624	80.49	67.35	0.7933
Component 2	88.56	79.46	0.8798	88.08	78.70	0.8750
Component 3	88.99	80.16	0.8842	89.70	81.33	0.8928
Component 7	90.43	82.53	0.8992	91.14	83.72	0.9068
The effectiveness of RIEM						
Component 1	86.89	76.83	0.8624	80.49	67.35	0.7933
Component 5	88.28	79.02	0.8770	89.29	80.65	0.8885
Component 2	88.56	79.46	0.8798	88.08	78.70	0.8750
Component 6	89.71	81.34	0.8918	90.50	82.65	0.9000
The effectiveness of BT						
Component 3	88.99	80.16	0.8842	89.70	81.33	0.8928
Component 5	88.28	79.02	0.8770	89.29	80.65	0.8885
Component 1	86.89	76.83	0.8624	80.49	67.35	0.7933
Component 4	83.88	72.24	0.8310	77.73	63.57	0.7691

In addition to the weak augmentation strategy, we incorporate a strong augmentation strategy consisting of ColorJitter, RandomGrayscale, Blur, and Cutout. Fig. 8 illustrates the effects of these data augmentation techniques.

D. Ablation Studies

In order to further investigate the contributions of each module in RISL, we conduct ablation experiments in this section. As shown in Tables V and VI, seven ablation experiments are conducted on the LEVIR-CD and WHU-CD datasets using 20%/5% labeled data and 20%/5% labeled data, respectively.

1) *Component 1:* Supervise training of labeled data using only the primary CD network.
2) *Component 2:* On the basis of CD network, the result of adding CR module.
3) *Component 3:* On the basis of CD network, the result of adding RIEM and BT module.
4) *Component 4:* On the basis of CD network, the result of adding BT module.
5) *Component 5:* On the basis of CD network, the result of adding RIEM module.

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TABLE VI

RISL'S ABLATION RESULTS ON THE 5% LABELED LEVIR-CD AND WHU-CD DATASETS. EXCEPT FOR KAPPA, ALL SCORES ARE EXPRESSED IN PERCENTAGE (%)

Method	LEVIR-CD			WHU-CD		
	F1	IOU	Kappa	F1	IOU	Kappa
The effectiveness of CR						
Component 1	65.03	48.18	0.6367	70.79	54.79	0.6886
Component 2	84.53	73.20	0.8376	87.56	77.88	0.8707
Component 3	85.10	74.06	0.8433	87.04	77.06	0.8653
Component 7	90.01	81.84	0.8949	89.80	81.48	0.8932
The effectiveness of RIEM						
Component 1	65.03	48.18	0.6367	70.79	54.79	0.6886
Component 5	82.08	69.60	0.8121	85.87	75.24	0.8525
Component 2	84.53	73.20	0.8376	87.56	77.88	0.8707
Component 6	89.15	80.43	0.8858	89.10	80.34	0.8864
The effectiveness of BT						
Component 3	85.10	74.06	0.8433	87.04	77.06	0.8653
Component 5	82.08	69.60	0.8121	85.87	75.24	0.8525
Component 1	65.03	48.18	0.6367	70.79	54.79	0.6886
Component 4	76.67	62.16	0.7553	78.06	64.02	0.7715

6) *Component 6:* On the basis of CD network, the result of adding CR and RIEM module.
7) *Component 7:* On the basis of CD network, the result of adding CR, BT, and RIEM module.

1) The Effectiveness of CR: The CR extracts rich semantic information from unlabeled images while ensuring consistency across different views of the same image. To verify the effectiveness of CR, we conducted two sets comparative ablation experiments (Components 1 and 2, Components 3 and 7). From the two sets of experiments, it is evident that CR significantly improves the performance of the CD network.

2) The Effectiveness of RIEM: The RIEM aims to evaluate the reliability of the entire instance sample from the perspective of individual objects. It integrates the representation information of individual objects, such as color, shape, and structure, to address the shortcomings of single-threshold methods that lack contextual correlation. The comparative ablation experiments of Components 1 and 5 show that RIEM can effectively enhance the performance of CD networks. In addition, the comparative ablation experiments of Components 2 and 6 have demonstrated that RIEM can further improve the model's performance based on CR.

3) The Effectiveness of BT: The BT is designed to provide more reliable and independent instance objects for RIEM. It effectively trims the edges of each instance object, preventing connectivity between different types of instances. The ablation comparative experiments of Components 3 and 5 show that BT can effectively promote the performance of CD networks with RIEM. However, we found that using BT alone in a CD network can suppress the network's performance (Components 1 and 4). This situation is because although low-threshold BT can effectively trim the boundaries of instances, it preserves too many false positive samples.

To explore the impact of data augmentation, we have added a data augmentation ablation experiment, as shown in Table VII.

TABLE VII

DATA AUGMENTATION ABLATION EXPERIMENTS USING DIFFERENT STRONG DATA ON 10% LABELED LEVIR-CD DATASET. WA IS WEAK AUGMENTATION STRATEGY. SA IS STRONG AUGMENTATION STRATEG. EXCEPT FOR KAPPA, ALL SCORES ARE EXPRESSED IN PERCENTAGE (%)

Data Augmentation	F1	IOU	Kappa
Original	83.99	72.39	0.8321
ColorJitter+WA	87.42*/86.59**	77.65*/76.34**	0.8678*/0.8592**
Grayscale+WA	87.87*/87.25**	78.37*/77.39**	0.8725*/0.8661**
Blur+WA	86.19*/87.18**	75.73*/77.28**	0.8548*/0.8654**
CutOut+WA	87.31*/87.82**	77.48*/78.29**	0.8666*/0.8720**
SA+WA	90.32*/89.82**	82.34*/81.51**	0.8981*/0.8929**

In Table VII, original represents the result of training using only labeled data. * indicates that the probability parameter for strong data augmentation is 0.5. ** indicates that the probability parameter for strong data augmentation is 0.7. Due to the CR method requiring weak and strong data augmentation methods, we fixed the weak augmentation strategy unchanged and only changed the strong augmentation strategy. Four different strong augmentation strategy experiments were conducted namely ColorJitter, Grayscale, Blur, and CutOut. We found that the Grayscale augmentation strategy has the most significant performance improvement on CD networks. And when the parameter is 0.5, the CD network is trained better than when the parameter is 0.7.

E. Experimental Comparison

To evaluate the superiority of the proposed RISL, we compare it with five state-of-the-art SSCD models, including AdvEnt, S4GAN, SemiCDNet, SemiCD, and UniMatch, on the WHU-CD, LEVIR-CD and CDD-CD datasets.

1) *AdvEnt [58]:* An approach for unsupervised domain adaptation in the semantic segmentation domain is based on adversarial entropy minimization. It utilizes entropy loss and adversarial loss to address the domain adaptation task in pixel-wise prediction-based semantic segmentation with loss of information.
2) *S4GAN [59]:* A dual-branch semi-supervised semantic segmentation method is employed, which leverages two network branches to establish a connection between semi-supervised classification and semi-supervised segmentation.
3) *SemiCDNet [22]:* A semi-supervised learning method for CD is proposed, consisting of dual-shared networks and dual-branch networks. The dual-shared networks aim to leverage the full potential of the data, while the dual-branch networks are designed to distinguish the types of labeled data and eliminate interference among different types of data.
4) *SemiCD [23]:* An incremental SSCD method is introduced, effectively utilizing unlabeled RS images for training through two alignment strategies.
5) *UniMatch [24]:* A holistic unified dual-stream perturbation method is proposed. It introduces an auxiliary feature perturbation stream to expand the perturbation space. A dual-stream perturbation technique is

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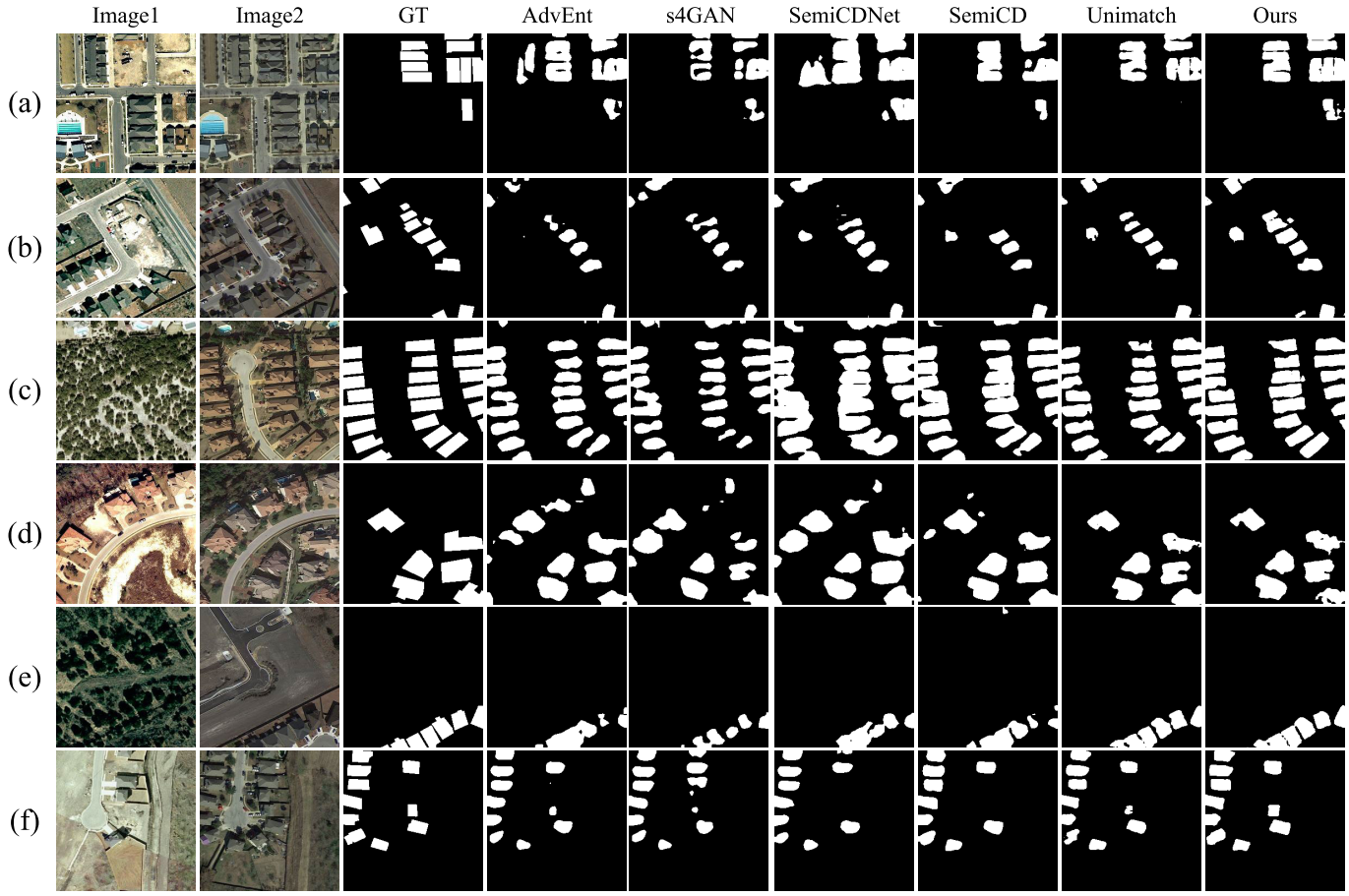


Fig. 9. Predicted results of different models on the LEVIR-CD dataset with 5% labeled data. For each row, from left to right are the prechange image, postchange image, Ground Truth, AdvEnt model prediction, s4GAN model prediction, SemiCDNet model prediction, SemiCD model prediction, UniMatch model prediction, and RISL prediction, respectively. (a)–(f) Represents representative samples selected from the LEVIR-CD dataset.

introduced, which enables two views to be simultaneously guided by a standard weak view.

AdvEnt and S4GAN are GAN-based SSCD methods. SemiCDNet is an SSCD method specifically designed for change detection tasks. SemiCD and UniMatch are consistency-based methods that leverage the concept of consistency for training.

To ensure fairness, we set the backbone of all methods as ResNet-50. Table II shows the results of these methods on WHU-CD, CDD-CD. and LEVIR-CD datasets. The table shows that our proposed RISL can achieve better performance than other methods in a small amount of labeled data. For example, on the 5% labeled CDD-CD dataset, it outperforms SemiCDNet method by 10.82 points in $F1$, 15.6 points in IOU, 0.1223 points in Kappa, and 16.61 points in Rec.

Compared with other consistency-based methods, such as SemiCD and UniMatch, RISL still performs well on three experimental datasets. On the 10% labeled LEVIR-CD dataset, RISL is 0.37 points higher than UniMatch on $F1$ and 3.24 points higher than SemiCD on $F1$.

Compared with other GAN-based methods AdvEnt and S4GAN, RISL achieved better performance results on various distributed labeled data. For example, On the 40% labeled WHU-CD dataset, RISL is 8.7 points higher on IOU than AdvEnt and 10.37 points higher on IOU than S4GAN.

In addition, in the LEVIR-CD and CDD-CD datasets under four different label distributions, the RISL method is much higher than the Only-sup method (the only-supervised methods, i.e., only use labeled data for training under different labeled ratios) and even higher than the Full-sup method (fully-supervised methods, i.e., the oracle method). In the WHU-CD dataset, although RISL's performance is not better than the Full-sup method, it is still much higher than the Only-sup method. This result indicates that the RIEM, boundary pruning module (BT), and CR module proposed by RISL can deeply mine rich and practical information in the dataset, enhancing the generalization ability of the CD network.

As shown in Fig. 5, RISL effectively mines semantic information hidden in unlabeled data. When the number of labeled data is small, but the number of unlabeled data is large, RISL performs better than other methods.

In order to further demonstrate the superiority of the proposed RISL, we calculated the number of parameters, training time, testing time, and FLOPs for RISL and five comparison methods, as shown in Table VIII. Table VIII shows that although RISL has a high number of parameters, its training time is lower than UniMatch and only higher than AdvEnt and SemiCDNet. During testing, RISL has lower parameters, FLOPs, and testing time than the other five comparison methods. Overall, RISL has the shortest testing time, the

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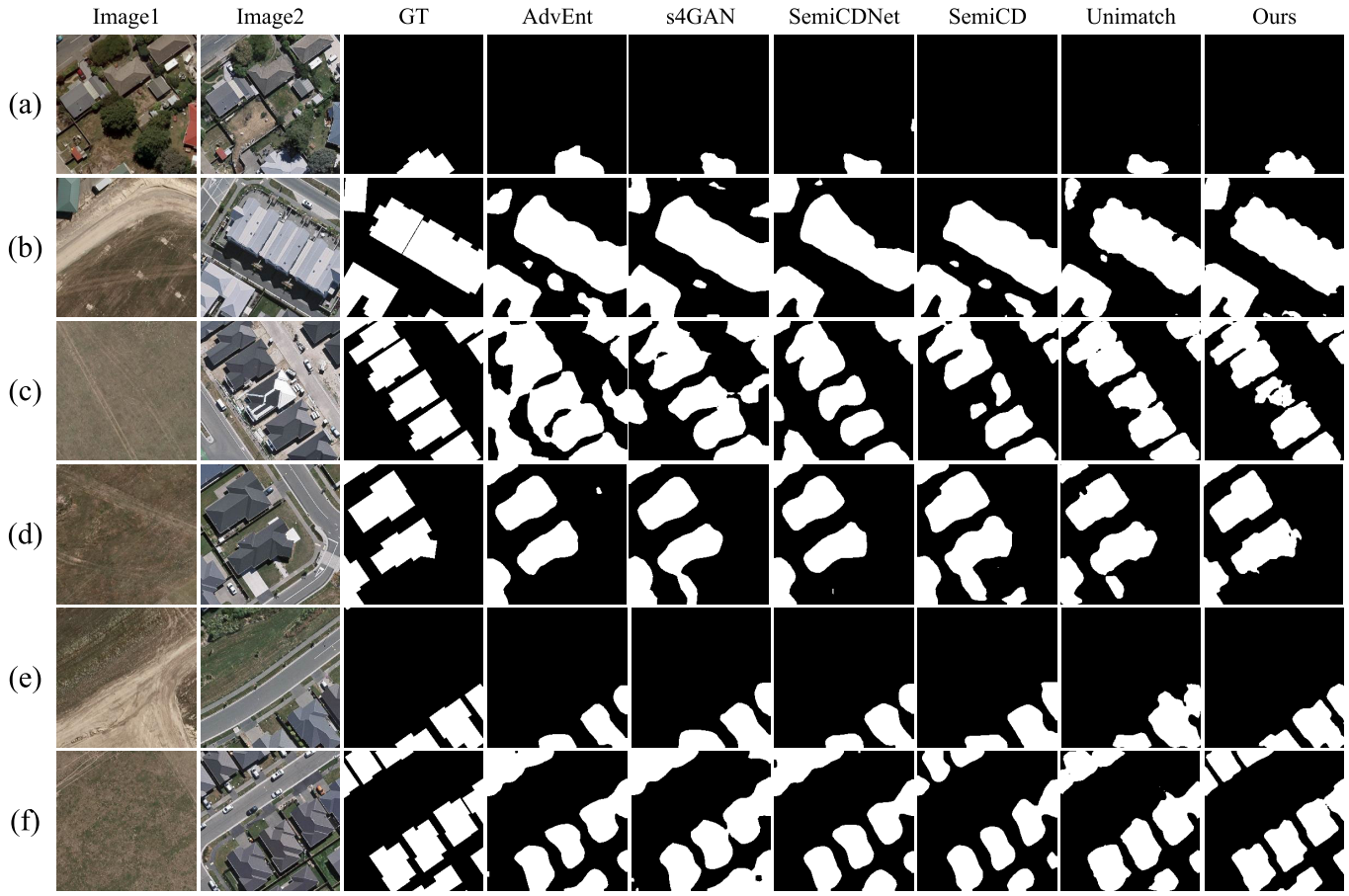


Fig. 10. Predicted results of different models on the WHU-CD dataset with 20% labeled data. For each row, from left to right are the prechange image, postchange image, Ground Truth, AdvEnt model prediction, s4GAN model prediction, SemiCDNet model prediction, SemiCD model prediction, UniMatch model prediction, and RISL prediction, respectively. (a)–(f) Represents representative samples selected from the WHU-CD dataset.

TABLE VIII
COMPARISON OF PARAMETERS, FLOPS, AND CONSUMPTION TIME (THE BATCH SIZE IS 8) OF DIFFERENT CD METHODS DURING TRAINING/TESTING

Method	Train Parameters(M)	Train FLOPs(G)	Train Time(s)
AdvEnt(Resnet-50)	49.62	148.11	201.70
S4GAN(Resnet-50)	49.62	148.23	235.11
SemiCDNet(Resnet-50)	52.40	153.00	140.13
SemiCD(Resnet-50)	50.70	151.80	259.54
UniMatch(Resnet-50)	40.47	163.89	244.04
RISL(Resnet-50)	51.65	64.23	227.55
Method	**Test Parameters(M)**	**Test FLOPs(G)**	**Test Time(s)**
AdvEnt(Resnet-50)	46.85	73.23	12.01
S4GAN(Resnet-50)	46.85	73.23	14.41
SemiCDNet(Resnet-50)	46.85	73.23	12.40
SemiCD(Resnet-50)	46.85	73.23	13.22
UniMatch(Resnet-50)	40.47	29.10	4.94
RISL(Resnet-50)	40.47	29.10	4.84

lowest FLOPs, and the least number of parameters used during testing. Although it is not the shortest regarding training parameters and time, it has achieved better performance.

F. Visualization Result

To visually compare the differences between RISL and other methods, we have selected representative results from the LEVIR-CD and WHU-CD dataset for demonstration purposes.

Fig. 9 displays the prediction results trained on the LEVIR-CD dataset with 5% labeled. The performance of our proposed RISL on LTPs is demonstrated in the (a)th row and (b)th row. For LTPs, the STFP would mistakenly filter out this portion of the samples. In contrast, the RIEM method evaluates the samples based on the entire instance object, which can effectively retain this part of the samples. The (c)th row and (d)th row demonstrate that RISL can effectively filter

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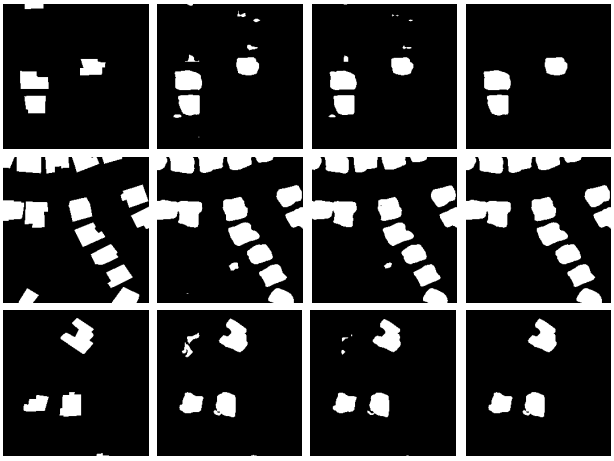


Fig. 11. Visualization results of the ablation experiments. (a) Ground Truth. (b) Pseudolabels were obtained from the CD network. (c) Pseudolabels were obtained from the BT. (d) Pseudolabels were obtained from the RIEM.

out HFPs. This demonstrates the effectiveness of evaluating sample reliability based on the entire instance object when dealing with a less robust CD network trained with limited labeled data, as it can efficiently filter out confusing prediction results. From the (a)th, (c)th, and (e)th rows, it can be observed that RISL can refine the edges of each object. In the (f)th row, it can be seen that RISL can both prevent the omission of LTPs and avoid the false detection of HFPs.

Fig. 10 displays the prediction results trained on the WHU-CD dataset with 20% labeled. Similarly, the (b)th row, (c)th row and (d)th row demonstrate the effective filtering ability of RISL for HFP. Lines (a)th row, (e)th row and (f)th row further demonstrate that RISL avoids excessive cropping of LTP edges and preserves edge information.

In Fig. 11, we visualize the pseudolabels generated after processing with BT and RIEM, demonstrating the effectiveness of the modules intuitively. It can be observed that after applying BT, the pseudolabels directly predicted by the CD network effectively filter out the LFPs, resulting in higher-quality pseudolabels. After applying RIEM, more HFPs are effectively filtered out, significantly improving the quality of the pseudolabels. These robust pseudolabels expand the training data, making the CD network more robust.

V. Conclusion

This article proposes a RISL change detection for RS images. First, this article conducts a detailed study of the pseudolabels generated by existing SSCD models. The experiment finds that the STFP cannot effectively filter out HFPs under high-confidence threshold and erroneously filters out LTPs. We argue this is due to the lack of semantic contextual information in STFP. To solve this problem, RISL combines the contextual semantic information of each instance object to evaluate the reliability of the instance, thereby obtaining robust pseudolabels. RISL consists of three modules, namely RIEM, BT, and CR. The RIEM is constructed to evaluate the reliability of the entire instance object. Compared to STFP, it overcomes the lack of semantic information in STFP. It determines the reliability of the instance by combining the semantic information and contextual relationships of the entire instance object. BT is designed to perform preliminary processing on the prediction of change detection models to minimize LFP and avoid class confusion in each instance object. Meanwhile, to utilize unlabeled training samples more effectively, we use consistent regularization (CR) to mine rich semantic information from unlabeled data in different views. RISL achieves state-of-the-art results on three widely used change detection datasets: LEVIR-CD, WHU-CD, and CDD-CD.

Although this work performs well on sub-meter spatial resolution datasets, there are often potential challenges and limitations when using low-resolution data (e.g., spatial resolution of 5 m).

1) *Spatial Resolution Limit:* A resolution of 5 m means that each pixel represents 5 m on the actual ground × A 5-m area. Such data may not be sufficient to capture small-scale changes or details, such as changes in small buildings or roads. The CD network is also more difficult to extract detailed information from it.
2) *Mixed Pixel Problem:* In boundary areas or transition zones, a pixel may contain multiple types of ground objects, which leads to a mixed pixel problem and affects the accuracy of change detection.

These two challenges are present in RISL, and other change detection models have to face them as well. For these two challenges, we can use models pretrained on high-resolution data and then fine-tuned on low-resolution data on migration learning, which can help the models adapt to the new data distribution quickly. Therefore, migration from high resolution to low resolution is a worthy direction for research.

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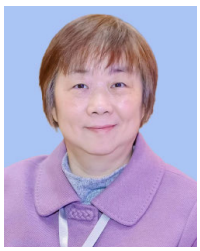
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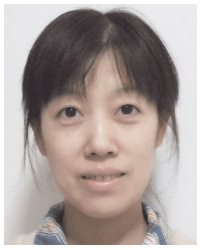
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